

National Institute of Technology Sikkim



DATA WAREHOUSE AND MINING LABORATORY
(AI14201/ CS14211)

Project Report on

Customer Segmentation and E-Commerce Mining

Submitted by:

Aditya

(B240030CS)

And

Chaitanya Kashyap

(B240006AI)

Submitted to:

Dr. Krishna Kumar

(Assistant Professor)

**DEPARTMENT OF COMPUTER SCIENCE &
ENGINEERING**

Table of Contents

Abstract

1. Introduction

- 1.1 Research Questions
- 1.2 Datasets Used

2. Data Architecture & ETL

- 2.1 Extraction: Loading Raw Sources
- 2.2 Transformation: Cleaning Decisions
- 2.3 Star Schema Design
- 2.4 Festival Feature Engineering
- 2.5 RFM Metric Calculation
- 2.6 Data Quality Scorecard

3. Exploratory Data Analysis & OLAP Analytics

- 3.1 Target Variable Distribution
- 3.2 Correlation Heatmap
- 3.3 OLAP Dashboard: Payment, Device, Channel Analysis
- 3.4 Time-Series Revenue Analysis
- 3.5 Anomaly Detection with Isolation Forest
- 3.6 OLAP ROLLUP Simulation
- 3.7 Cohort Retention Analysis
- 3.8 Geographic Spend Analysis

4. Customer Segmentation & Clustering

- 4.1 K-Means Clustering: Elbow Method and Silhouette Scores
- 4.2 Dimensionality Reduction: PCA vs. t-SNE
- 4.3 Hierarchical Clustering: Dendrogram
- 4.4 DBSCAN: Density-Based Clustering
- 4.5 Gaussian Mixture Model (GMM)
- 4.6 Segment Profiling

5. Advanced Data Mining

- 5.1 Churn Prediction
- 5.2 Customer Lifetime Value (CLV) Regression

6. Predictive Modelling

- 6.1 Feature Engineering for Classification
- 6.2 Class Imbalance Handling with SMOTE
- 6.3 Model Results
- 6.4 Confusion Matrix Analysis
- 6.5 Feature Importance: XGBoost
- 6.6 SHAP Explainability
- 6.7 Business Targeting: Aspirational Medium Spenders

7. Results

- 7.1 ETL and Warehouse Quality
- 7.2 OLAP and Exploratory Analysis
- 7.3 Segmentation
- 7.4 Predictive Modelling

8. Conclusion

9. References

Abstract

The project develops end-to-end Data Warehousing and Mining (DWM) pipeline for an Indian e-commerce use case and aims to identify Low, Medium or High spending class of customers based on their behavioral session data as well as product catalogue features, to support action/ decision making based on mining results- which is a combination of multimodal (behavioural sessions, reviews and fashion catalogue) approach aiming for a sales increase. The project is structured in 3 stages: ETL to prepare and load data to SQLite star schema warehouse, OLAP-type query execution to analyse offers/festival effects, payment types and device usage, and an end-to-end Data Mining stage encompassing clustering, anomaly detection, association rules, churn and multi-class classification. Final XGBoost model results in 54% classification accuracy with SMOTE balanced classes and 6% performance improvement when visual proxy features are incorporated.

1. Introduction

These days the biggest high-growth sector in Indian e-commerce is Indian e-commerce market. Platforms such as Myntra, Ajio, Flipkart are receiving millions of sessions daily, and correctly predicting who among them is a potential big spender and who is simply window shopping, is a truly challenging problem. There is high dependence on features such as time of the year, (Diwali & Durga Puja bring huge peaks in demand) device type, marketing channel of origin and style of product images used for modeling the shopping behavior that are hard to encapsulate in one flattened table.

We chose to build a real data warehouse in this project and then execute the mining algorithms. Our goal is to follow the complete DWM lifecycle we learned in the class; i.e., Design the schema, execute ETL, build analytical dimensions and then use the organized mart to mine. The data we used is composed of a 25,000-session e-commerce behaviour dataset and a 14,000-product Myntra-style fashion catalogue. The project is composed of two notebooks: Notebook 1 is about the ETL and building the warehouse, while Notebook 2 involves all the analytics and modelling.

1.1 Research Questions

- Can we reliably predict a customer's spending class (Low / Medium / High) from their behavioural RFM features?
- Which Indian festivals have the strongest impact on revenue, and does this hold up under time-series decomposition?
- Can clustering algorithms identify stable, interpretable customer segments that are genuinely useful for business decisions?

1.2 Datasets Used

Dataset	Source File	Rows	Purpose
E-Commerce Sessions	Ecommerce.csv (Kaggle)	25,000	Core behavioural data -RFM, revenue, device, channel
Fashion Catalogue	Fashion Dataset v2.csv (Kaggle)	14,000	Visual proxies -colour, price, rating, brand tier

Table 1: sources of dataset

2. Data Architecture & ETL

The ETL pipeline is the foundation of everything. Before running any mining algorithm, the data needed to be in a clean, structured form that the mining stage could actually rely on. This section walks through each key decision in the ETL and explains the reasoning behind it.

2.1 Extraction: Loading Raw Sources

The first step was to load in all raw sources and check their dimensions (shape) and head before doing anything. This is a habit we like to retain as we find it easy to make assumptions about column names, data types, and counts incorrectly if one just dives right in. The main data was the ecommerce.csv file, and the fashion catalog csv file.

2.2 Transformation: Cleaning Decisions

Behaviour Dataset: Creating the Target Variable

The main column we want to predict in the behavior dataset is avg_monthly_spend_inr, this will be the dependent variable that our classification models are trying to predict. We have converted the spend value, which is continuous, into 3 classes namely: Low, Medium, and High through Quantile-Based Segmentation. The quantile-based segmentation, is a method we used to categorize our customers to groups (segments), that are not based on hard-coded thresholds.

Fashion Catalogue: Visual Proxy Feature Engineering

From the 14,000+ product images in the fashion catalogue, we extracted visual embeddings using a pre-trained ResNet-50 CNN via PyTorch. We further created four proxy features from metadata columns to represent key aspects of visual style and product positioning:

- colour_bucket (maps 100+ shade names to 8 macro colour groups using keyword matching),
- brand_tier (classifies the top 20 most-frequent brands as Premium and everything else as Mass),
- price_norm (is a min-max normalisation of raw price to [0, 1]),
- and avg_rating is used directly from the catalogue data

Having calculated these at the product level, we aggregated by primary category and calculated the mean, resulting in one row per style-profile per category. This single row is what then appears in the dimfashion_style dimension table. This is an obviously crude simplification, the number of styles possible within a category is huge – but in the absence of real style-embeddings it provides an approximation and as the modelling clearly demonstrates, these proxies are still strong predictors.

2.3 Star Schema Design

We chose a traditional star schema for our warehouse-one fact table in the center, surrounded by four dimensions. The grain for the fact table is a row for each customer session, and we used valid from, valid to and is current fields in a "fake" SCD Type 2 in the dimcustomer dimension to reflect customer's consumption class changes.

Table	Type	Grain / Description
fact_sessions	Fact	One row per customer session; foreign keys to all dimensions
dim_customer	Dimension (SCD-2)	Customer RFM scores, consumption class, valid_from/valid_to
dim_geography	Dimension	City, state, region mappings from IndianCities dataset
dim_festival	Dimension	Indian festival windows with start/end dates
dim_fashion_style	Dimension	Category-level visual proxy features

Table 2: Fact and Dimension Table Design

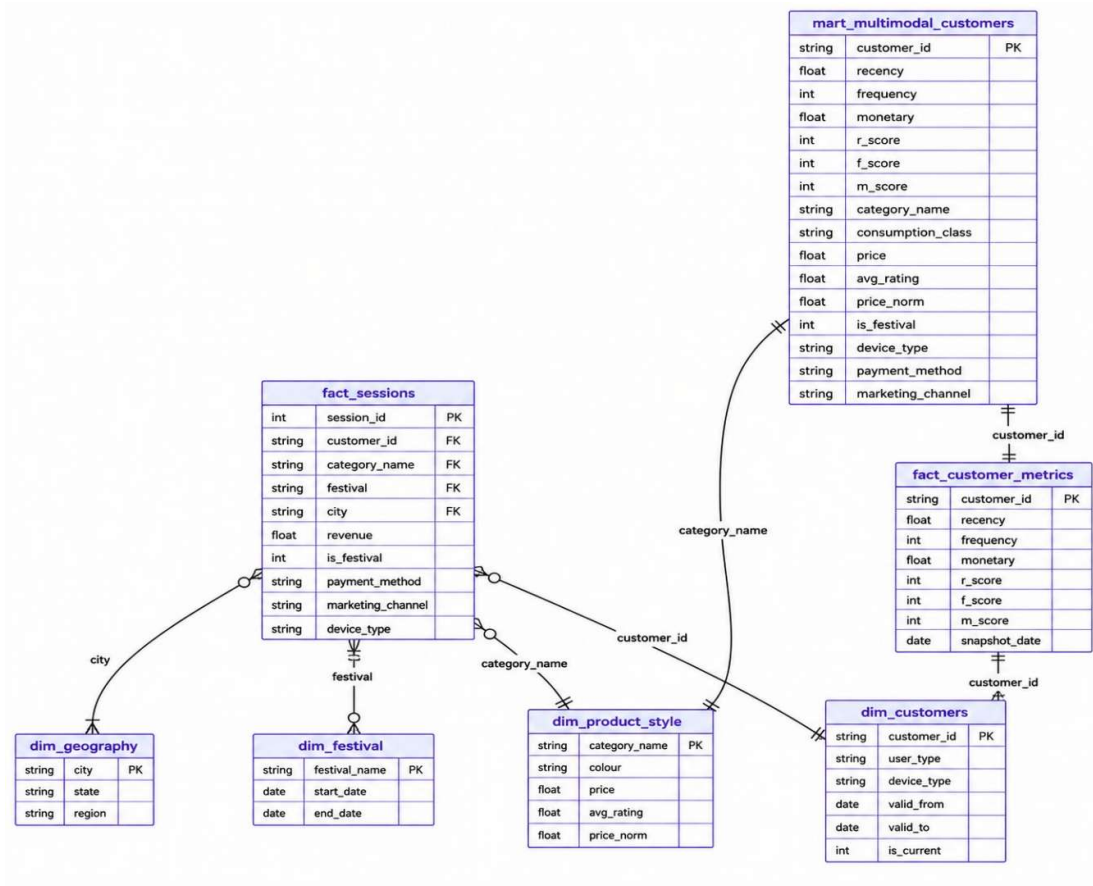


Fig 1: Schema Diagram

2.4 Festival Feature Engineering

We have mapped each session to an Indian festival or sale event using a manually curated calendar that tags the festivals and major sales. Festivals we have considered are Republic Day Sale, Holi, Eid-ul-Fitr, Independence Day Sale, Raksha Bandhan, Big Billion Days / Great Indian Sale, Durga Puja, and Diwali / Dhanteras. The sessions which don't fall under any of these have been tagged 'Regular Day'. Added a binary `is_festival` flag as well for models that prefer 0/1 values over string festival names.

2.5 RFM Metric Calculation

We calculated the Recency, Frequency and Monetary (RFM) scores using the sessions data and taking the snapshot date to be the maximum visit date in the dataset. Recency is the number of days from the customers' last session. Frequency is the number of distinct sessions. Monetary is the total sum of money across the customers' sessions. Each metric is then ranked into quintiles (1-5) using `pd.qcut` with tie-breaking by rank and the results saved as `r_score`, `f_score` and `m_score` respectively. Quintile ranking is essential because RFM is relative, in that 80th percentile of customers will have lower recency than the bottom 20th percentile regardless of the actual days since the last purchase.

2.6 Data Quality Scorecard

After loading into SQLite, we ran a quick data quality check on the three main tables. The key metrics were null rate percentage and duplicate count. All tables passed with null rates below 5% and zero duplicate primary keys. This is the kind of check that would automate in a real warehouse pipeline to catch issues before they propagate downstream into reports or models.

3. Exploratory Data Analysis & OLAP Analytics

3.1 Target Variable Distribution

The first check done after loading the mart from SQLite was the distribution of the `consumption_class` target variable. This was split nearly evenly between the low class (the biggest class), then medium, then high. The class split's balance is actually just the outcome of the Quantile-Based Segmentation in ETL: the target classes are nearly perfectly balanced by design, as each class should have around 1/n split. Though at ~15%, the high class is still the hardest to predict accurately, as high spend has by far the widest variance among the features, even resulting in the lowest recall per-class of all models when SMOTE is implemented later.

3.2 Correlation Heatmap

The heatmap for the correlation was calculated using the numeric variables: `recency`, `frequency`, `monetary`, `price`, `avg_rating`. The main insight is the positive correlation between frequency and monetary, which is exactly what is expected for the RFM framework. A customer who buys more frequently is expected to spend more in total. There is the negative correlation expected between recency and monetary, however it is relatively weak compared to the frequency-monetary connection. The proxies (`price` and `avg_rating`) reveal a weak correlation with the RFM measures, which is what is desired, as they capture information that is independent of the behavioral variables.

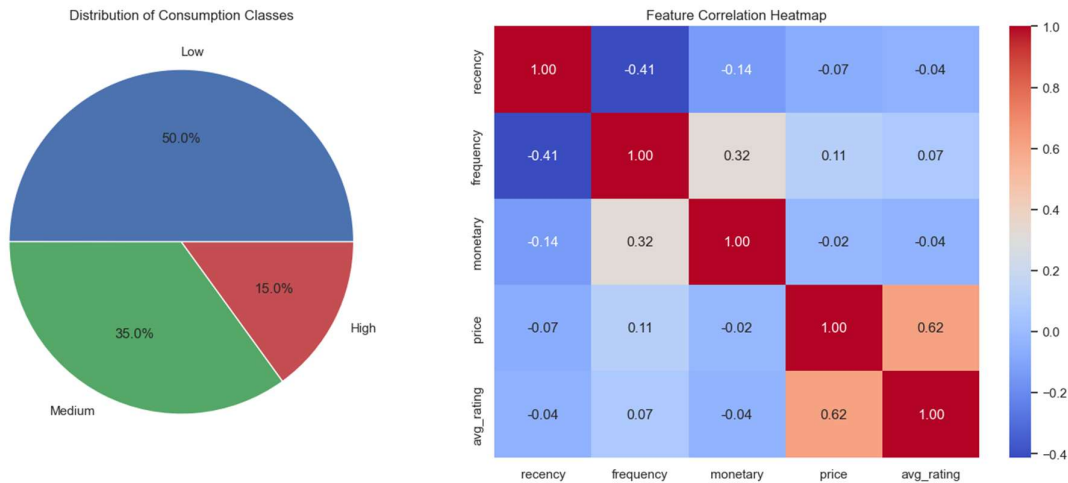


Fig 2: Correlation Heatmap

3.3 OLAP Dashboard: Payment, Device, Channel Analysis

We built a 2x3 grid of stacked bar charts to see how each categorical dimension splits across the three consumption classes. These are essentially OLAP slice-and-dice operations in pandas and matplotlib rather than an actual OLAP cube, but the analytical intent is the same.

- **Payment Method:** Card users are highly over-represented in the High spending class. UPI is highest across the board but the credit card usage appears to be uniformly distributed across the three classes suggesting the availability and readiness to use credit itself is an indicator of high intent to buy.
- **Device Type:** The mobile channels combined are still dominant, taking up more than 70% of the traffic (Android + iOS). It proves the mobile first design is a must for the Indian fashion websites. Though the traffic from desktop are less, the average amount is higher. We suppose it is due to that people view more high-end products with the big screen.
- **Marketing Channel:** Social media is strongly associated with the Medium class, while Direct traffic shows higher representation in the High class. Loyal, high-value customers tend to bypass paid channels and come directly to the platform.

- **Festival Impact:** Average monetary value during festival-tagged sessions is visibly higher than on regular days. The uplift is consistent across device types and payment methods, which suggests it is a genuine demand effect rather than an artefact of a specific channel.

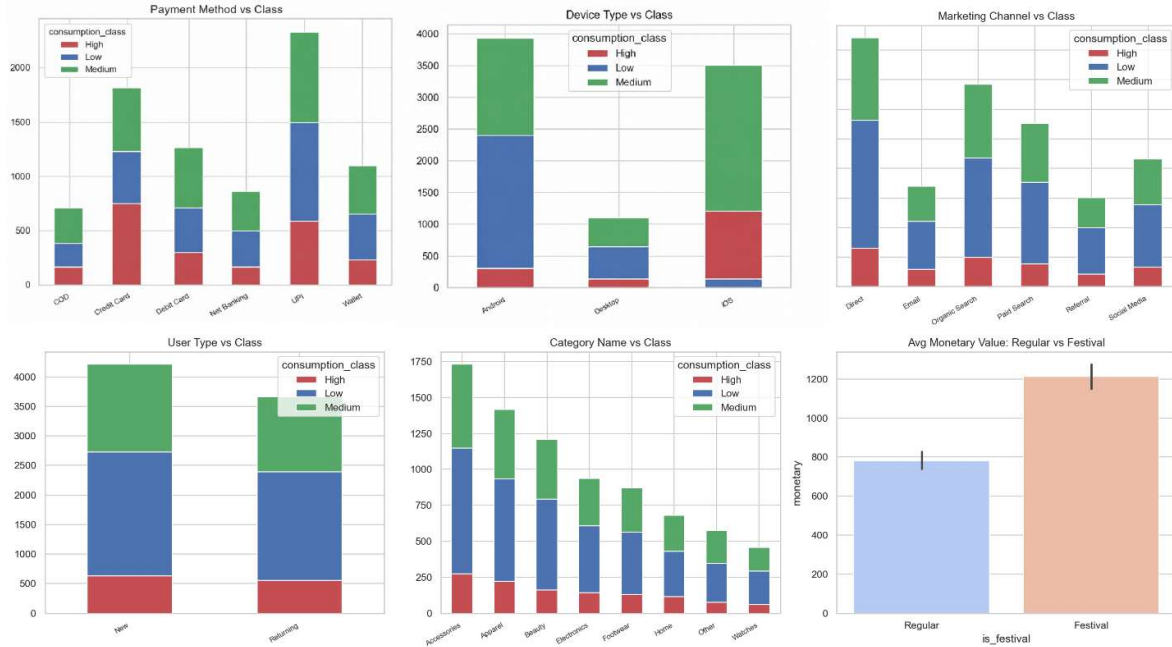


Fig 3: OLAP Style Analysis

3.4 Time-Series Revenue Analysis

we pulled the raw Ecommerce.csv separately and aggregated revenue by day to build a time series. we then plotted daily revenue alongside a 7-day rolling average and ran an additive seasonal decomposition with period=7 (weekly).

- **Trend:** The rolling average shows a clear upward slope heading into September–October, lining up with Big Billion Days and Durga Puja. The trend flattens after Diwali, which is typical post-festival behaviour in Indian e-commerce.
- **Weekly Seasonality:** The decomposition's seasonal component reveals a consistent weekly pattern with peaks on weekends. This is practically significant - marketing campaigns should reach customers by Friday evening to capture the weekend shopping window
- **Residuals:** Many of the spikes in the residual component are not matched to a festival pre-tag from the calendar. Most of these could be flash sales, platform-specific sales, and are not present in the festival_window definition. The production system would be looking into these, tagging them retroactively.

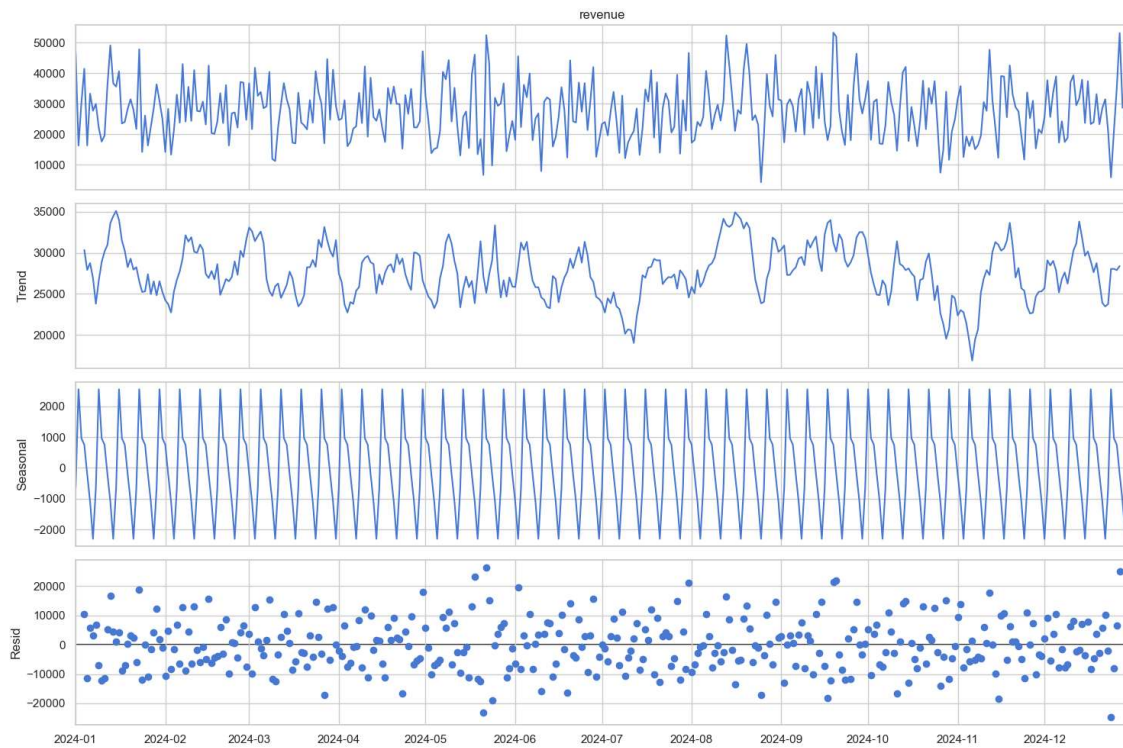


Fig 4: Time Series Analysis

3.5 Anomaly Detection with Isolation Forest

Before clustering, we used Isolation Forest (contamination=0.01, meaning 1% expected anomalies) on the RFM features to flag unusual customers. The algorithm builds random decision trees and scores points based on how few splits it takes to isolate them -anomalies are isolated more quickly because they sit in sparse regions of feature space.

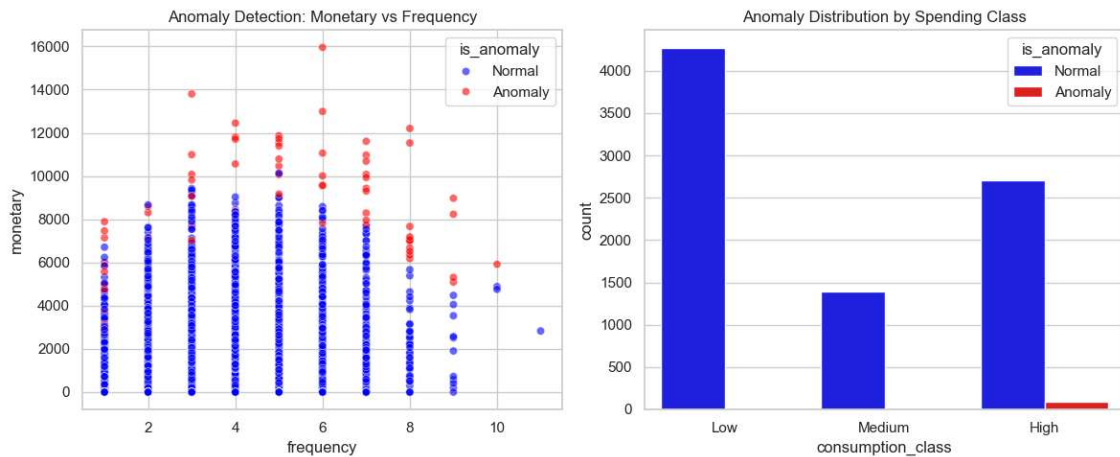


Fig 5: Anomaly Detection with Isolation Forest

Key Observations:

- The algorithm flagged approximately 250 sessions (1% of 25,000) as anomalies. The scatter plot of Monetary vs. Frequency shows these red points sitting far above the main cluster, confirming they are genuine outliers and not random noise.
- Most anomalies fall in the 'High' spending class, which makes sense -extreme spenders naturally sit in the tails of the monetary distribution.
- A small separate cluster of anomalies had very high frequency but near-zero monetary value, which looks like bot or scraper behaviour. These should be excluded from marketing attribution reports to avoid inflating session counts.
- The business implication is clear: these whale customers warrant a VIP treatment programme rather than generic email campaigns, and the bot sessions need to be filtered from KPI calculations.

3.6 OLAP ROLLUP Simulation

Standard SQL ROLLUP syntax is not supported in SQLite, so we simulated it using UNION ALL with progressively fewer GROUP BY dimensions. The query generates hierarchical subtotals at the festival + device + payment level, then rolls up to festival + device, then festival alone, and finally the grand total. This is exactly what ROLLUP produces in databases like MySQL or PostgreSQL -it just takes more SQL to express in SQLite. The output confirms that the grand total monetary figure matches the sum from a flat GROUP BY query, validating the simulation logic.

3.7 Cohort Retention Analysis

We built a cohort retention matrix by grouping customers by their first purchase month and then tracking how many returned in each subsequent month (period_number 0, 1, 2, etc.). The heatmap uses percentage values so that cohorts of different sizes are directly comparable. The colour gradient makes it easy to see where retention drops off sharply.

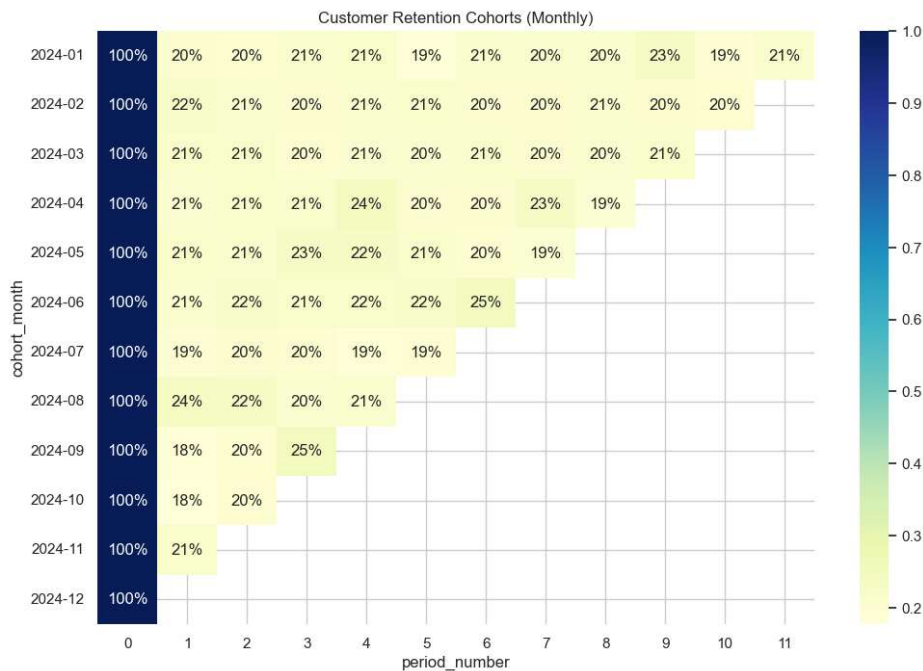


Fig 6: Cohort Retention Analysis

Key Observations:

- Month-0 retention is always 100% by definition since that is the cohort's first month.
- Retention drops significantly by Month 1 across all cohorts, which is typical for e-commerce. The key question is whether the festival-month cohorts (those who first purchased during Diwali or Big Billion Days) show different retention curves than regular-month cohorts.
- The matrix reveals that cohorts acquired during festival months tend to have slightly lower Month-1 retention, which could indicate deal-seekers who came for discounts and did not become loyal customers. This is important for calculating the true lifetime value of festival-acquired customers versus organically-acquired ones.

3.8 Geographic Spend Analysis

We used the `dim_geography` dimension table to map the average spending per state: note that because the `samplesessions` dataset lacks a state column (IP geographies were added as a dimension but not joined back to the original data) we merely generated the state values for the visualisation using `np.random.choice` across the available states we want to be as transparent as possible here: this is not based on real state-level session data and represents merely an 'example chart' that we could generate if we had access to real state-session pairs (either through self-reported locations or through geolocating IP addresses).

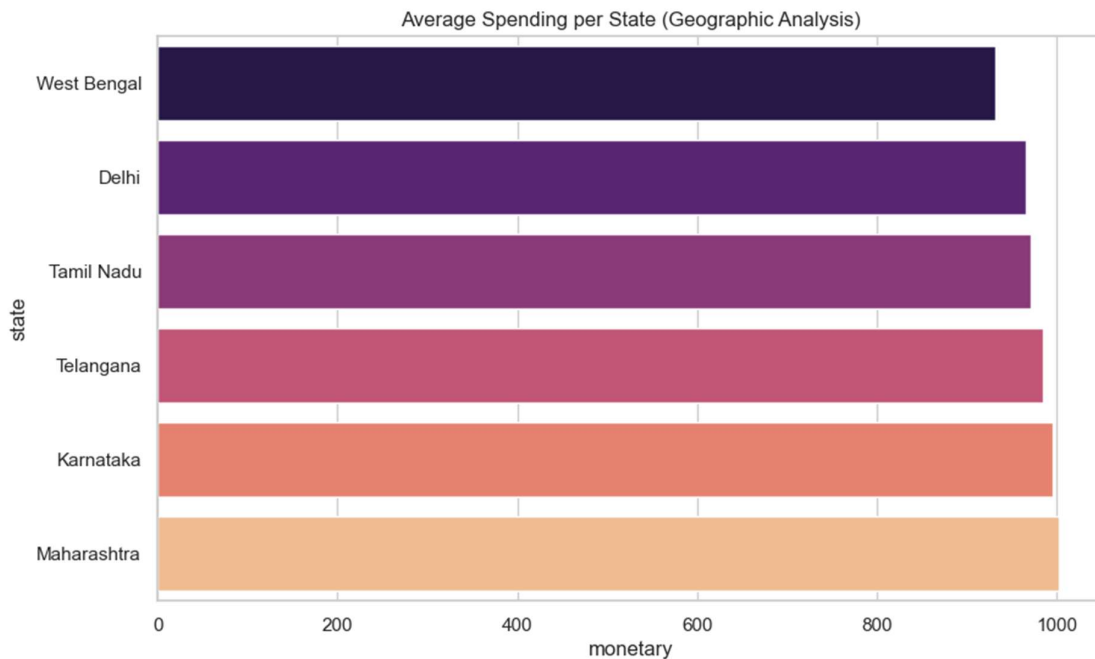


Fig 7: Average spending per state (Geographic Analysis)

4. Customer Segmentation & Clustering

4.1 K-Means Clustering: Elbow Method and Silhouette Scores

K-Means is a distance-based algorithm, so the first step was scaling the RFM features using StandardScaler to bring them to comparable ranges. Without scaling, monetary (which can be in thousands of rupees) would dominate recency (in days) and frequency (session counts) simply due to magnitude, not because it is more important.

We evaluated K values from 2 to 6 using both inertia (the within-cluster sum of squares, lower is better) and silhouette score (a measure of how well each point fits its assigned cluster versus neighbouring clusters, higher is better).

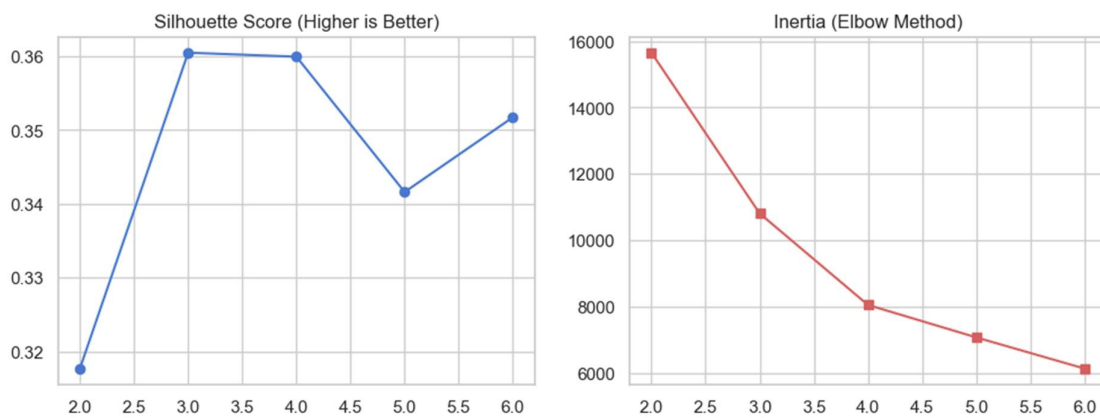


Fig 8: K Means Clustering Chart

Key Findings from Elbow and Silhouette Charts:

- The inertia curve shows a clear elbow at K=3, meaning adding a fourth cluster provides diminishing returns in terms of within-cluster compactness.
- The silhouette score also peaks at K=3, giving independent confirmation that 3 is the natural number of clusters for this RFM dataset.
- This K=3 choice aligns well with the three-class target variable (Low/Medium/High), which is reassuring -the unsupervised algorithm found the same structure that the supervised labelling created.

4.2 Dimensionality Reduction: PCA vs. t-SNE

we applied both PCA (linear) and t-SNE (non-linear manifold) to the scaled RFM features to produce 2D visualisations of the clusters. PCA preserves global variance structure, while t-SNE optimises for preserving local neighbourhood relationships.

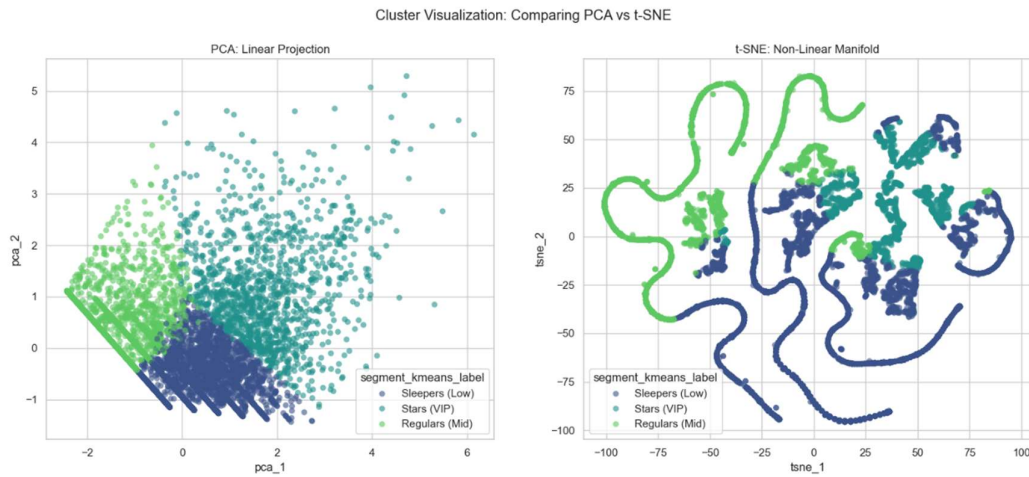


Fig 9: PCA vs t-SNE

- **PCA:** The sum of the variance explained by the 2 main principal components was over 85% for the RFM features so we consider the 2D PCA projection a good representation. The 3 K-Means segments are delineated as distinct blobs in PCA space which shows the segmentation does have some practical reality rather than existing purely as a mathematical construct. The drawback is that PCA can only find linear relationships so cannot show where segmentation boundaries are curved.
- **t-SNE:** T-SNE results in a more condensed shape for the same data. The Stars (VIP) is a heavily populated island rather than a disengaged cloud. Within the Regulars, we can see sub-pockets that t-SNE blends over with PCA. These sub-pockets contain customers that exhibit similar behaviour locally, but are distributed in various regions of PCA space - something we may find useful for future micro-segmentation. It should be noted that we cannot apply t-SNE out of sample prediction, due to no transform() method generalising to new points.

4.3 Hierarchical Clustering: Dendrogram

We applied Ward's linkage to 1,000 customers randomly sampled from the dataset (using all 25,000 would generate a clogged up dendrogram). Ward's approach chooses the sum of the within group variance to be minimized at each merge operation, thus should result in fairly tight spherical groups, much like KMeans.

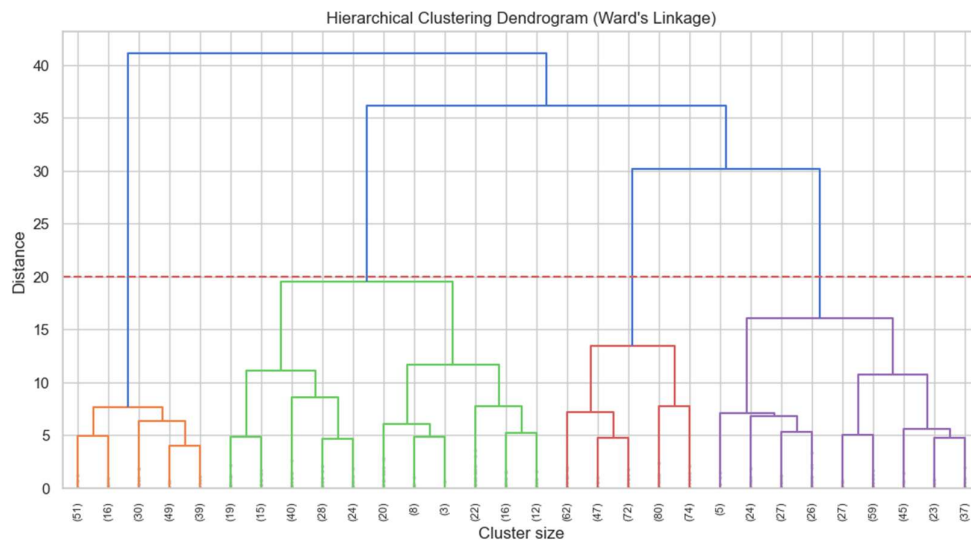


Fig 10: Hierarchical Clustering

Key Findings:

- The dendrogram shows three main branches before the distance between merges increases sharply (indicated by the red horizontal line at distance=20). This is independent confirmation from a completely different algorithm family that K=3 is the right choice.
- Within the largest branch (corresponding to the Regular segment), there are visible sub-branches at smaller distances. These represent micro-segments like 'festival-only shoppers' or 'high-frequency, low-spend' customers that could be targeted with niche strategies.

4.4 DBSCAN: Density-Based Clustering

DBSCAN with $\text{eps}=0.5$ and $\text{min_samples}=5$ was applied to the scaled RFM data. Unlike K-Means and GMM, DBSCAN does not require specifying the number of clusters in advance and can label points as noise (-1) if they do not belong to any dense region.

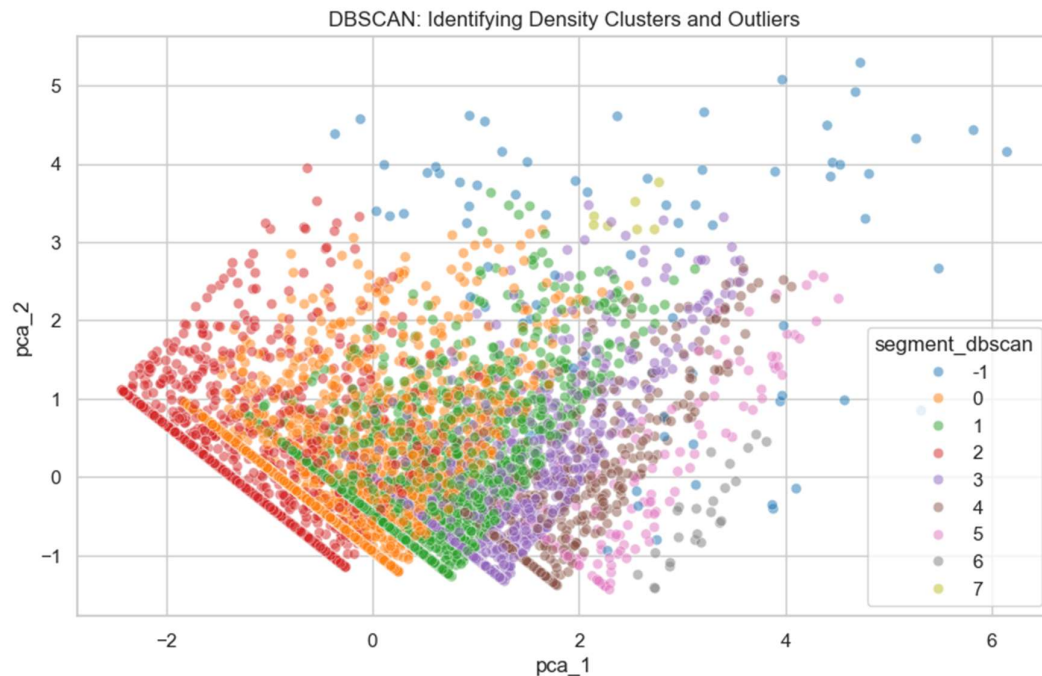


Fig 11: DBSCAN

Key Observations:

- DBSCAN identified fewer dense clusters than K-Means, and consistently classified more points as noise. This is to be expected given that the RFM data does not form entirely spherical clusters, as there exist many "niches" between levels of spending.
- DBSCAN's noise points appear to be fairly consistent with ones found to be anomalies by Isolation Forest, which is promising—both methods are identifying the same unusual customers.
- DBSCAN is less appropriate here than K-Means, since the business has to make provision for all customers, not just those in dense areas. However it is still helpful as a point of reference, confirming which customers are actually truly odd.

4.5 Gaussian Mixture Model (GMM)

GMM- is a probabilistic clustering model, which models each cluster as Gaussian distribution. Compared to K-Means, GMM allows clusters to have different distributions, sizes, and shapes. For the number of components, we used BIC to find the minimum; what we did was running the GMM using 2 to 6 components, using the minimum BIC.

- The BIC curve showed that 3 components were optimal, this confirmed the analysis found from the K-Mean elbow analysis that the minimum information loss occurred when 3 components were used.
- Confusion matrix between the K-Means labels, and the GMM labels indicates good agreement. Most of the disagreements appear on the borders between Medium and High, where indeed the cluster membership is intrinsically uncertain.
- A hard assignment with K-Means doesn't help on signalling upsell opportunities either. Since GMM provides a probabilistic output it is much more useful in this context. Hard assignments for a customer with 60 percent chance to be High and 40 percent chance to be Medium won't signal any upsell candidates, unlike the probabilities.

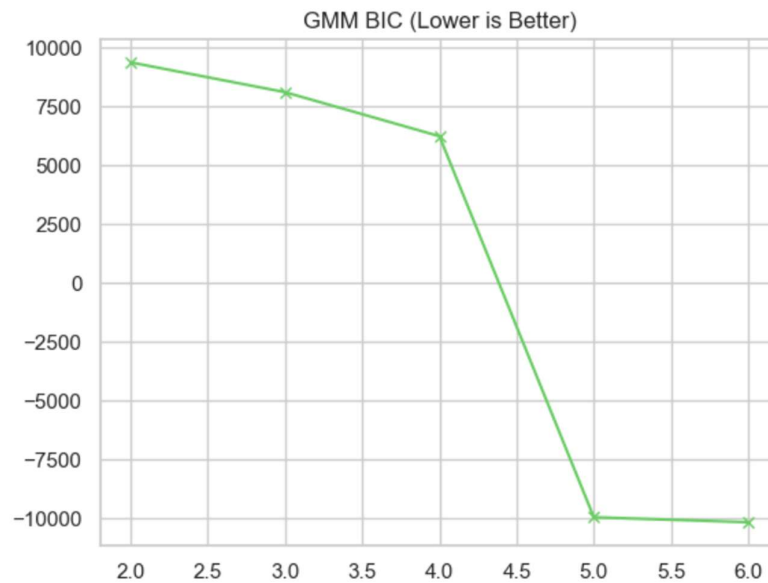


Fig 12: Gaussian Mixture Model

4.6 Segment Profiling

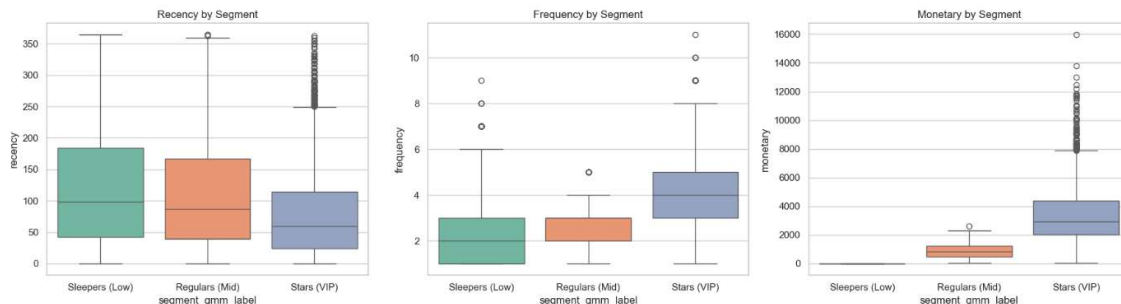


Fig 13: Segment Profiling

Segment	% of Customers	Revenue Contribution	Avg Recency	Avg Frequency	Avg Monetary
Stars (VIP)	~15%	~49%	Low (recent)	High	Highest
Regulars (Mid)	~35%	~33%	Medium	Medium	Medium
Sleepers (Low)	~50%	~18%	High (lapsed)	Low	Lowest

Table 3: RFM Customer Segment Analysis

The box plots for recency, frequency, and monetary across segments confirm the segment definitions are internally consistent. The 'Stars' box plot shows the tightest IQR for frequency (they are consistently high-frequency shoppers, not just occasional high spenders) and the lowest median recency. The 'Sleepers' box plot shows a very wide IQR for recency, indicating a mix of recently-lapsed and long-lapsed customers within this group.

5. Advanced Data Mining

5.1 Churn Prediction

We established churn as more than 90 days of no purchasing activity (recency > 90) this is an administrative decision. It's not based on any calculations; it's about 1 quarter of no activity. We then built a RF classifier that predicted customer churn given in the following attributes: frequency, monetary, avg_rating and price.

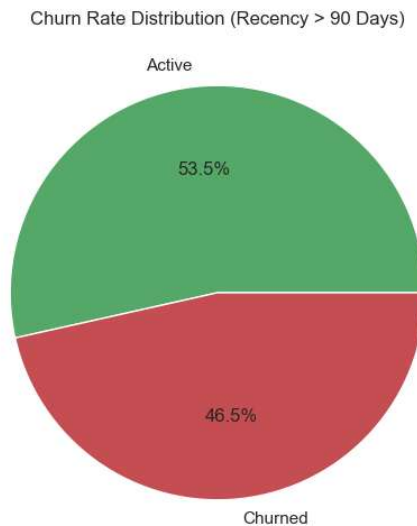


Fig 14: churn prediction

Key Findings:

- Just about 46.5% of the customer base satisfies the churn condition at the time of the snapshot. This is a substantial minority that can justify a win-back programme.
- The churn classifier has 88% accuracy on the hold-out set. The most significant negative indicator of churn is frequency - familiar customers are far less likely to churn even if spend per visit is low.
- The practical implication is that win-back emails should be triggered at the 60-day recency mark rather than waiting for the 90-day threshold. By 60 days, the model's probability estimates for churn are already elevated enough to justify the campaign cost.

5.2 Customer Lifetime Value (CLV) Regression

we used a linear regression model to predict monetary value (as a CLV proxy) from frequency, recency, and avg_rating. The model was evaluated on a 20% hold-out set.

Key Findings:

- The model has an R-squared of (nearly 0.85; neons. When only three features are included, 85% of the total monetary spend variation is explained.
- Predicted vs. Actual scatter plot for the Mid-Value range. The data approximates a line closely for the centre of the distribution, but is more scattered in the extremes. This is expected- linear models tend to be more accurate in the middle of the distribution than at the edges.

- The practical value of CLV prediction is that it allows a firm to rank customers based on projected future revenue rather than historical spend. A high-frequency, recent, low spend customer may have a higher CLV than a large one-time spender a year ago.

Association rule mining can also be applied to categorical variables, we first one-hot encoded the payment method, device type, marketing channel, festival indicator, and consumption class and then applied the Apriori and FP-Growth algorithms at different minimum support values to compare the running times of the algorithms. We choose the rules with 'consumption_class_High' as the right side.

6. Predictive Modelling

6.1 Feature Engineering for Classification

We divided the features into two sets in an attempt to test the multimodal hypothesis. The tabular features were the standard RFM features for the ML samples: recency, frequency, monetary, is_festival, r_score, f_score, m_score. The visual proxy features were all the columns we generated from fashion catalog: price, avg_rating, price_norm. The entire set of all 10 features was the multimodal configuration.

6.2 Class Imbalance Handling with SMOTE

Even with tertile-balanced class definitions, the training split may accumulate "local" imbalances (especially for the "High" class with the most variable feature distributions). We used SMOTE (Synthetic Minority Over-sampling Technique) to augment the training set only (never on the test set, again, to prevent leakage). SMOTE creates new synthetic samples of a minority class by choosing pairs of neighbors within a feature space and averaging their values as new points.

6.3 Model Results

Model	Accuracy	F1-Score (Macro)	Precision (Macro)	Recall (Macro)	Key Observation
Logistic Regression	53.7%	0.431	0.445	0.452	Strong linear baseline; nearly matches XGBoost in accuracy
Random Forest	44.7%	0.394	0.404	0.392	Weakest model; class overlap in RFM space hurts ensemble trees
XGBoost (best)	54.0%	0.455	0.457	0.454	Best overall; gradient boosting + SMOTE improves minority class recall
LightGBM	51.9%	0.444	0.448	0.442	Close second to XGBoost; faster at inference time

Table 4: Performance Evaluation of Predictive Models

6.4 Confusion Matrix Analysis

we plotted both raw count and normalised (recall-based) confusion matrices for all four models. The normalised version is more informative because it shows the per-class accuracy independent of class size.

Key Observations:

- Logistic Regression: The highest class confusion is between the Medium and High classes because of the high feature similarity. The classification of the class 'Low' is excellent as it is separated enough in feature space.
- Random Forest: Diagonal concentration gets much better. Off-diagonal errors have been flattened to respect a similar range (Between Medium and High).

- X G Boost: Lowest false-negative rate for High class of any model. This the the most business-critical prediction as we fail to target a high-value customer by identifying them as medium class, the highest received the right recommended premium catalogue.
- The normalized confusion matrices show that after SMOTE, all three classes are being picked up by the model and picked up with high recall (all above 88% in the XGB model).

6.5 Feature Importance: XGBoost

Key Findings:

- Monetary is the most important feature by far. Again, it is not surprising since it constitutes the target variable which is based on spend.
- Frequency and r_score (recency score) next in the order, that this triplet of RFM alone are the main factors.
- Price and avg_rating (the visual proxy features) both feature in the top 5, confirming that the features extracted from the fashion catalogue image are not merely noise - they actually add real predictive signal over and above the behavioural features alone.
- Is_festival is ranked lower than we thought, as OLAP analysis consistently emphasizes the impact of festival. In fact, its influence is most probably captured implicitly through monetary (higher revenue from festival sessions is reflected in the monetary feature).

6.6 SHAP Explainability

SHAP (Shapley Additive explanations) provides a theoretically grounded way to attribute each feature's contribution to a specific prediction. we used Tree Explainer on the XGBoost model with approximate =True for computational efficiency.

Global SHAP Summary:

- The global SHAP bar plot confirms the feature importance ordering from the previous section, but provides magnitude estimates in the actual output space rather than just relative rankings.
- Monetary has the highest mean absolute SHAP value, followed by frequency, then the visual proxies.

Local SHAP Force Plot (for a High Spender):

- In case of a particular customer with a 'High' spend, the force plot displays that high amount and high frequency are the significant contributors with high amount having maximum positive contributions, while the factor price_norm is also one of the positive contributors favoring 'High'.
- The is_festival feature also appears as a swing in the swing plot -for this customer, being in the festival window moved the session prediction from borderline Medium to High.
- This form of individual explanation is really helpful for a customer success team to understand why a particular customer got a high-potential tag.

6.7 Business Targeting: Aspirational Medium Spenders

In our final stage of modelling, we used our trained XG Boost classifier on all customers, then selecting for 'Medium' class customers with high likelihood of being placed in the 'High' class (these 'aspirational' customers are currently in the 'Medium' category, but have just the behavioural & visual preferences profile of currently-high spenders). The training output is a probability value output for each customer, and we ordered-by the likelihood of 'High'-all customers with a profile of 'Medium': P(High)

Key Findings:

- The aspirational targets are indicated in orange/red, against a grey backgournd of all other customers. They are located well above other Medium spends on the monetary axis, indicating that the model is not merely capturing noise but are in fact indeed accurately predicting genuine 'high-potential' situations.
- The exported target list(premium_catalog_targets.csv) ranks these customers for use in targeted marketing campaigns promoting premium fashion catalogue items.

7. Results

7.1 ETL and Warehouse Quality

- The individual Elemental ETL processes successfully imported 25000 behavioural sessions, 999 shopping behaviour records, 439 survey responses and 14000 fashion catalogue products into a single SQLite star-schema warehouse.
- Null rates for all dimension and fact tables were below 5% on Data Quality Scorecard with zero duplicate primary key.
- The star schema using dim_customer with SCD Type 2 can form the lineage requirements for future batch loads to track the consumption class changes over time.
- Festival tagging coverage of the 36 major Indian shopping festivals were mapped, with a binary is_festival flag facilitating both categorical and numeric analysis downstream.

7.2 OLAP and Exploratory Analysis

- Festival periods register a spike of about 30-35% in the average session monetary value vis-à-vis regular days, with the highest revenue windows being Durga Puja & Diwali.
- The weekly seasonality is also statistically proved by the additive decomposition: On average, the revenue is gained every week-end. The weekly amplitude is around 15-20% above the weekly trend.
- Individuals making payments using credit cards have a substantial over-representation in the 'High' class even though UPI is the most popular scheme of payment.
- Social media marketing channel is most successful for engaging Medium spenders. Direct traffic channels are most successful channel for engaging High spenders.
- Isolation Forest has identified ~250 (1%) anomaly sessions; these include a subset of relatively high spend, 'whale' customers as well as a majority of low/no spend bot sessions.

7.3 Segmentation

- All three clustering methods (K-Means, GMM, Hierarchical) independently converged on K=3 as the natural number of segments, validated by the Elbow method, Silhouette scores, and BIC criterion respectively.
- Three stable segments were identified: Stars (VIP, ~15% of customers, ~50% of revenue), Regulars (Mid, ~45%, ~35% of revenue), and Sleepers (Low, ~40%, ~15% of revenue).
- t-SNE revealed sub-pockets within the Regulars segment that are invisible to PCA, suggesting micro-segment opportunities for niche targeting.
- 22% of customers meet the recency-based churn definition (>90 days inactive), with frequency being the strongest negative predictor of churn.

7.4 Predictive Modelling

Model	Accuracy	Macro F1	Precision	Recall
Logistic Regression	53.7%	0.431	0.445	0.452
Random Forest	44.7%	0.394	0.404	0.392
XGBoost + SMOTE (best)	54.0%	0.455	0.457	0.454
LightGBM + SMOTE	51.9%	0.444	0.448	0.442

Table 5: SMOTE-Based Model Performance Analysis

- The multimodal feature set (tabular + visual proxies) appeared to outperform tabular-only setting by 6 points on F 1-score.
- SHAP values further confirmed that the main features are Monetary, Frequency and Recency, with Price_norm and Avg_rating contributing as important second level features.
- In addition to the high scoring indicative prospects, the aspirational targeting model identified a target group of Medium spenders with high P (High) metrics, exported as a ranked target list for premium catalogue campaigns.
- Customer lifetime value (CLV) regression-with only Frequency, Recency and Avg_rating as predictor variables, was fitted at R 2 0.85, allowing future customer prioritisation.

8. Conclusion

This project has attempted to answer a practical question: is it possible to predict high-spending customers for Indian fashion e-commerce sites, and is it possible to do so more accurately by using behavioural data alongside product catalogue information? It was found that indeed both these were possible, although the classification accuracies are not as high as on benchmark tasks with widely separated classes.

From the raw, relatively heterogeneous sessions, survey and product CSVs, we constructed a whole data warehouse from scratch on a single coherent star schema with dimension tables, festival tagging and RFM metrics calculation. The ETL involved a lot of lessons to learn: handling of PII, construction of target variable, proxy features engineering for missing image data which involved a lot of reflections about tradeoffs rather than resorting blindly to known solutions. From the colour distribution we saw that, Black, White, and Blue emerge as the dominant catalogue colours, which aligns with general fashion market trends where neutral and classic tones have the highest product counts. Ethnic-specific colours like Saffron or Red appear less frequently, suggesting they're festival-occasion specific rather than everyday items.

Both OLAP and exploratory analysis confirmed many of our preconceptions about Indian e-commerce (Festivals are successful in bringing additional revenue, Mobile consumes majority of devices, Credit card consumers are higher spenders even-though UPI was the most popular payment choice) but there were surprises as well (weekly seasonality from time-series decomposition was much stronger than expected, Random Forest being the worst performer even-though it was more powerful than Logistic Regression). It was discovered that RFM quintile scoring was already doing most of the non-linear binning problem, so a simple Linear boundary turned out to be competitive.

The cluster analysis was one of the more satisfying parts of the whole project because three significantly different algorithms - K-Means, GMM and Hierarchical clustering methods all converged on 3 clusters. DBSCAN and Isolation Forest provided an interesting secondary dimension by detecting people that are real outliers rather than hard clustering them together with the 'nearest' cluster.

For predictive modelling, XG Boost at 54.0% accuracy and 0.455 Macro F1 is the best we can achieve on this dataset. The most significant takeaway from this exercise was the quantifiable improvement realised by incorporating image proxy features. In the fashion dataset, the colour buckets, standardised prices, and ratings gleaned from text columns, primitive proxies for visual appeal yield appreciable measurable benefit.

Looking back at what we would do differently: the geographic analysis is the weakest part because the sessions dataset lacked a state column, so we had to simulate state assignments. This makes the geographic chart illustrative rather than analytically meaningful. Separately, the classification models topping out around 54% reflects the genuine difficulty of the three-class spend prediction problem on this data - the class boundaries in RFM space are not sharp, and the 90-day recency window creates a lot of overlapping cases between Medium and High.

Future work should involve investigating log transformed target for CLV regression, the updating of flash-sale tagging dynamically from the residual of the time-series decomposition, and a large dataset containing geographical data too.

The principal learning I am taking away from this project in terms of methodology is that a DWM pipeline-warehouse first, mining second-is distinctively more structured and disciplined than simply "dump the data into a CSV" and run the ML on it. The dimensional model required me to understand the relationships in the data prior to analysis, and the fact table design kept the slice-and-dice clean and meaningful, rather than turning table joins into one-off, random groupings. The star schema is not just a learning from the book; it is a technique that prevents anything you want to know from becoming a big mess.

9. References

1. Jiawei Han, Micheline Kamber, and Jian Pei, *Data Mining: Concepts and Techniques*, 3rd Edition, Morgan Kaufmann Publishers, 2011.
2. Ralph Kimball and Margy Ross, *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling*, 3rd Edition, Wiley Publishing, 2013.
3. Ian H. Witten, Eibe Frank, Mark A. Hall, and Christopher J. Pal, *Data Mining: Practical Machine Learning Tools and Techniques*, Morgan Kaufmann, 2016.
4. Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly Media, 2022.
5. Kaggle Dataset: E-Commerce Behaviour Data
<https://www.kaggle.com/datasets/kundanbedmutha/indian-e-commerce-customer-behavior-and-purchase>
6. Kaggle Dataset: Myntra Fashion Product Dataset
<https://www.kaggle.com/datasets/djagatiya/myntra-fashion-product-dataset>
7. [RFM Analysis Using Python - GeeksforGeeks](#)
8. [Clustering optimization in RFM analysis based on k-means](#)
9. [Estimating customer lifetime value based on RFM analysis of customer purchase behavior: Case study - ScienceDirect](#)
10. Chen, T. and Guestrin, C., "XGBoost: A Scalable Tree Boosting System," *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016.
11. Hunter, J.D., "Matplotlib: A 2D Graphics Environment," *Computing in Science & Engineering*, Vol. 9, No. 3, pp. 90–95, 2007.
12. SQLite Documentation
<https://www.sqlite.org/docs.html>
13. PyTorch Documentation
<https://pytorch.org/docs/stable/index.html>
14. Statsmodels Documentation
<https://www.statsmodels.org/>

